



Q1. Why are filter capacitors used in some battery charger circuits across charging terminals and not in some others? What is the purpose of using a filter capacitor in a battery charger circuit?

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A1. The design of a battery charging circuit with or without filter capacitor depends on many factors including the type of battery and the charging method used. One must follow the safety precautions and have basic knowledge about the batteries being charged.

The most common battery charger circuit is the constant-voltage battery charger. It is basically a DC power supply that consists of a step-down transformer to reduce the AC mains voltage and a rectifier to provide the DC voltage to charge the battery.

Some chargers do not deliver pure DC and instead supply a pulsating (uneven) DC voltage. In such cases an output capacitor can be added across the circuit to smoothen the pulses.

In most of the common lead-acid battery charger circuits, capacitors are not required across the output terminals.

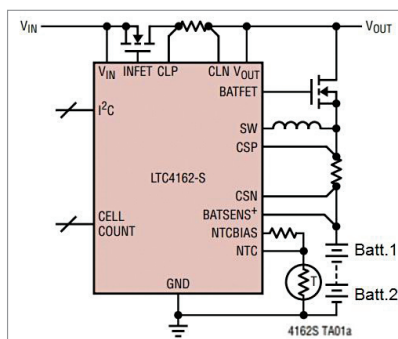


Fig. 1: Battery charger without output capacitor (Credit: analog.com)

However, charging a lithium-ion battery is slightly different and requires additional circuitry and a capacitor to protect the battery.

In some designs, a capacitor is used as a bypass capacitor across the output terminals to keep the control loop stable and to keep the

ripple voltage low when the battery is disconnected. The use of a bypass capacitor across the output terminals is shown in Fig. 2.

There are mainly two types of filter capacitors in battery charging circuits: input filter capacitor and output filter capacitor. The AC voltage across the step-down transformer is rectified and often filtered using capacitors to obtain a regulated DC voltage through a voltage regulator chip to charge the battery.

In most cases, the voltage regulator chip requires input capacitor at its input and also output capacitor at its output terminals. These input and output capacitors are located close to the chip. The input capacitor ensures that the voltage regulator chip connected across the transformer is stable and does not oscillate.

In a simple battery charging circuit, the battery is directly connected across the output capacitor of the voltage regulator chip. In most cases additional filter capacitors are not required in such circuits.

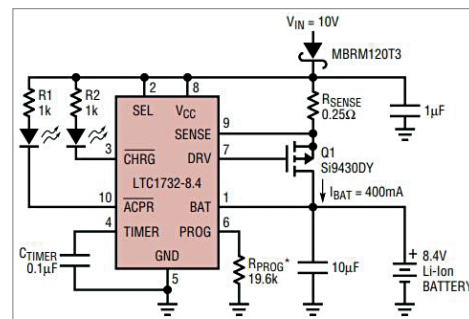


Fig. 2: Battery charger with bypass capacitor across output terminals (Credit: analog.com)

Q2. How does an inverter LED bulb work? How does it detect the switch state even during power cut?

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A2. An inverter LED bulb is essentially an emergency LED bulb. Actually, such products are somewhat confusing because there is no inversion process (DC to AC, or vice versa) involved in the circuits where these are used.

The technology used in these bulbs enables them to automatically turn on the LED light during a power cut and automatically charge the internal battery when AC mains is available. Basically, inverter LED bulb is a two-in-one bulb. That is, when there is AC mains power, it works as a normal light bulb but when there is a power cut, it works as an emergency light bulb.

Most inverter LED bulbs have lithium-ion batteries that provide three- to four-hour backup. These internal batteries require three to four hours of charging time to fully charge.

Thereafter the charging voltage is automatically cut to protect the battery from getting overcharged. The bulb is designed to charge the internal battery even during normal operation.

However, most inverter LED bulbs come with charge-only mode. This mode is used to charge the internal battery with the LED bulb off.

When AC mains power goes off, this inverter bulb automatically switches on to act as a DC emergency light. It can be switched off when not required.

The features of an inverter LED bulb include fully automatic control, low power consumption, and high efficiency. Since it has internal backup power source and electronic control circuit, it can detect the switch state even during power cut.